



Multiples of 10 involving negative exponents

Task A: Introducing negative exponents

1) Fill in the gaps:

	Decimal	Fraction	Exponential Form
a	0.09		
b		$\frac{4}{100\,000}$	
c			3×10^{-1}
d		$\frac{2}{10}$	
e	0.0007		
f			8×10^{-6}
g		$\frac{7}{1000}$	
h			5×10^{-4}

2) What mistake has Aisha made?

Write 0.0003 in expanded exponential form.

Aisha writes:

$0.0003 = 3 \times 10^{-3}$ because there are 3 zeros after the decimal point.



Aisha

3) Write each of the following in the alternative format:

	Decimal	Exponential form
a	0.34	
b		$5 \times 10^{-1} + 2 \times 10^{-2}$
c		$4 \times 10^{-2} + 7 \times 10^{-3} + 3 \times 10^{-4}$
d	0.049	
e	0.000603	
f		$2 \times 10^{-1} + 3 \times 10^{-4} + 3 \times 10^{-6}$
g	17.02	
h		$2 \times 10^2 + 3 \times 10^{-1} + 5 \times 10^{-3} + 8 \times 10^{-4}$



Task B: Writing integers in a range of forms

- 1) Identify the incorrect alternative in each row. Record the letter and rearrange them to make a word.

Number	Alternative 1	Alternative 2	Alternative 3	Alternative 4
3000	3×10^4 N	$3 \times 10^2 \times 10$ L	$\frac{30}{10} \times 10^3$ I	$\frac{300}{10} \times 10^2$ Y
74 000	7.4×10^4 A	74×10^3 P	7.4×10^3 X	$\frac{740\ 000}{10}$ C
120 000	$12 \times 10^2 \times 10^2$ R	1.2×10^5 H	1.2×10^4 E	$1.2 \times 10^6 \times 10^{-1}$ B
450	4500×0.1 S	$\frac{450}{10^3} \times 10^3$ E	4.5×10^2 F	4.5×10^{-2} O
8 000 000	$8 \times 10^3 \times 10^3$ T	$\frac{8}{10^3} \times 10^8$ P	8×10^6 V	$\frac{8000}{10^2} \times 10^5$ D
628 000	$6\ 280\ 000 \times 10^{-1}$ O	6.28×10^5 D	6.28×10^3 N	628×10^3 P
32 490	32.49×10^2 E	324.9×10^2 M	$\frac{3249}{10^4} \times 10^5$ A	$324\ 900 \times 10^{-1}$ H
12 400 000	$12.4 \times 10^{-3} \times 10^9$ N	$\frac{124}{10^2} \times 10^7$ L	1.24×10^7 O	$124 \times 10^{-2} \times 10^6$ T

